Sentence Processing

Could this be the sentence that launched a thousand experiments?

The horse raced past the barn fell.

What events are described in this sentence?

How could you disambiguate it?

The “garden path” effect.

The horse raced past the barn fell.

This sentence is almost as famous as “Colorless green ideas sleep furiously.”



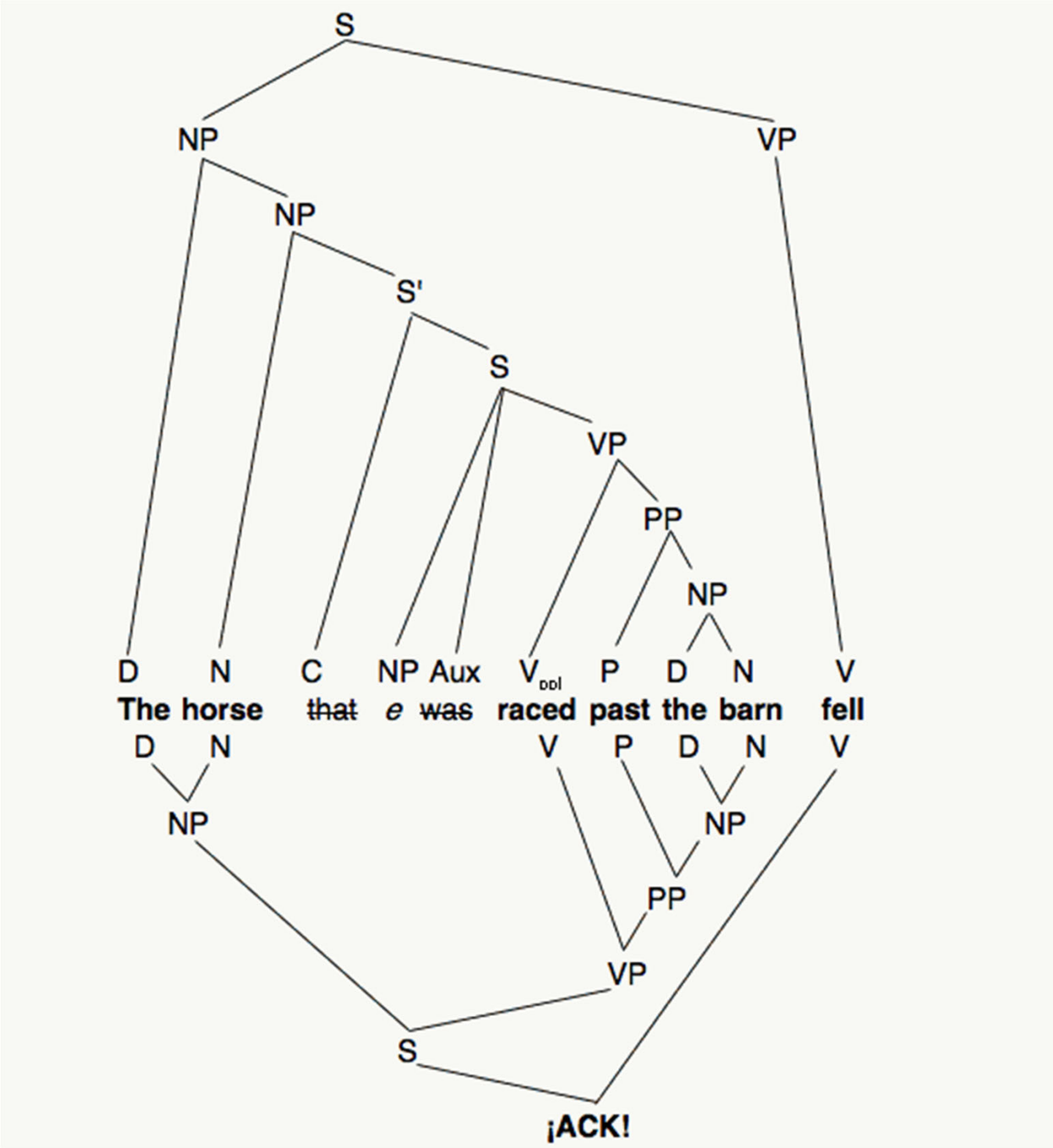
Syntactic explanations of the “garden path” effect

If we imagine that there is a parser whose job is to construct a syntactic tree for the sentences we hear/read, we have to have some way of dealing with the fact that it is implemented in a cognitive system with:

- limited memory
- time constraints

(these trees, and the dinosaur comic are both from the Laguage Log piece on this:

http://languagelog.ldc.upenn.edu/nll/?p=167)



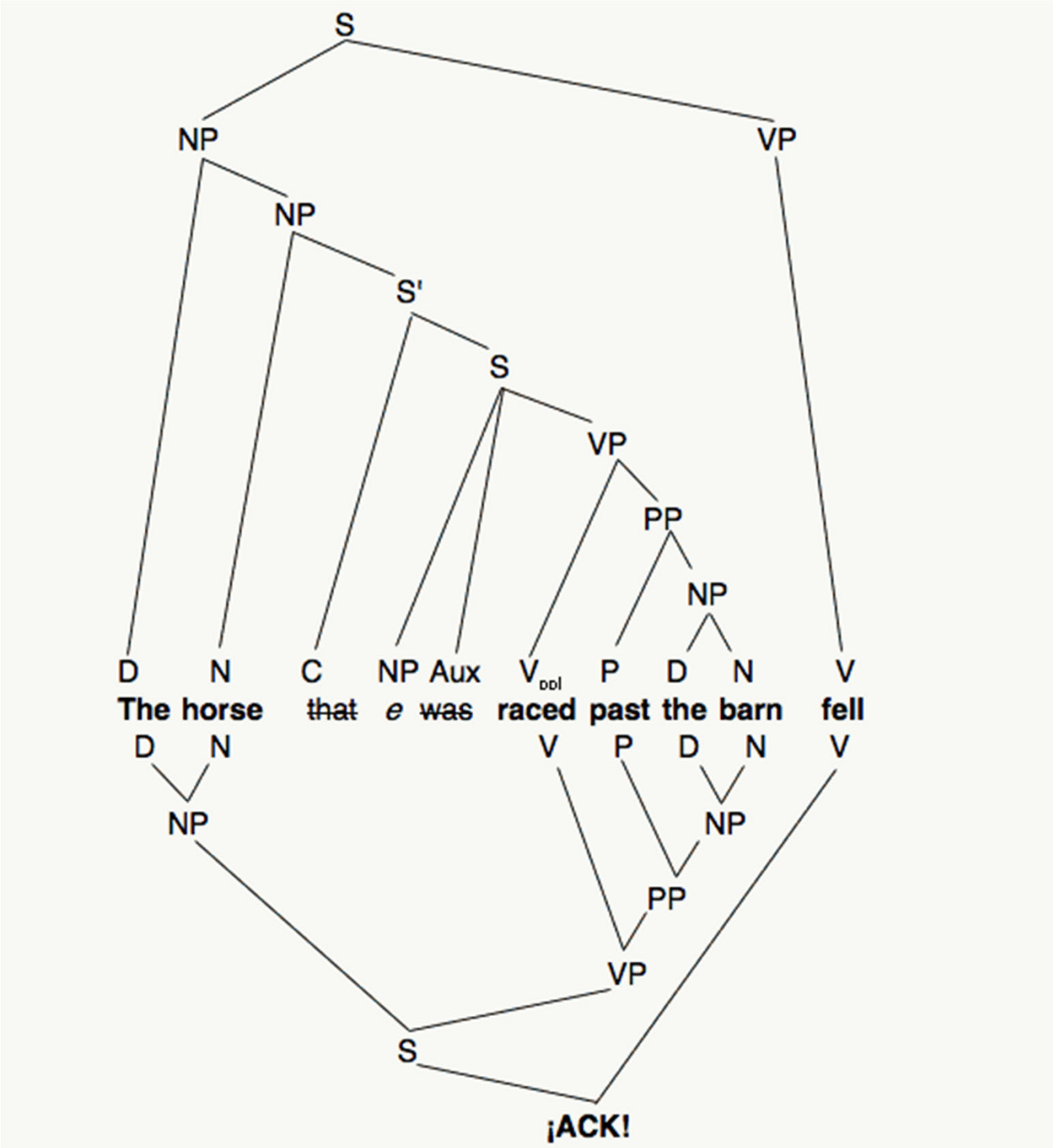
Syntactic explanations of the “garden path” effect

You can see that the grammatical version of this sentence is much more complex syntactically than the failed parse.

Specifically, it has many more nodes.

One strategy you might take to save space is to initially prefer smaller trees.

This is called “minimal attachment”



Syntactic explanations of the “garden path” effect

Another parsing strategy in this framework is “late closure.” The idea here is that there is a bias to keep the current clause “open” as long as possible.

In this case, “with the dynamite” wants to associate with the big verb phrase starting with “blew open.”

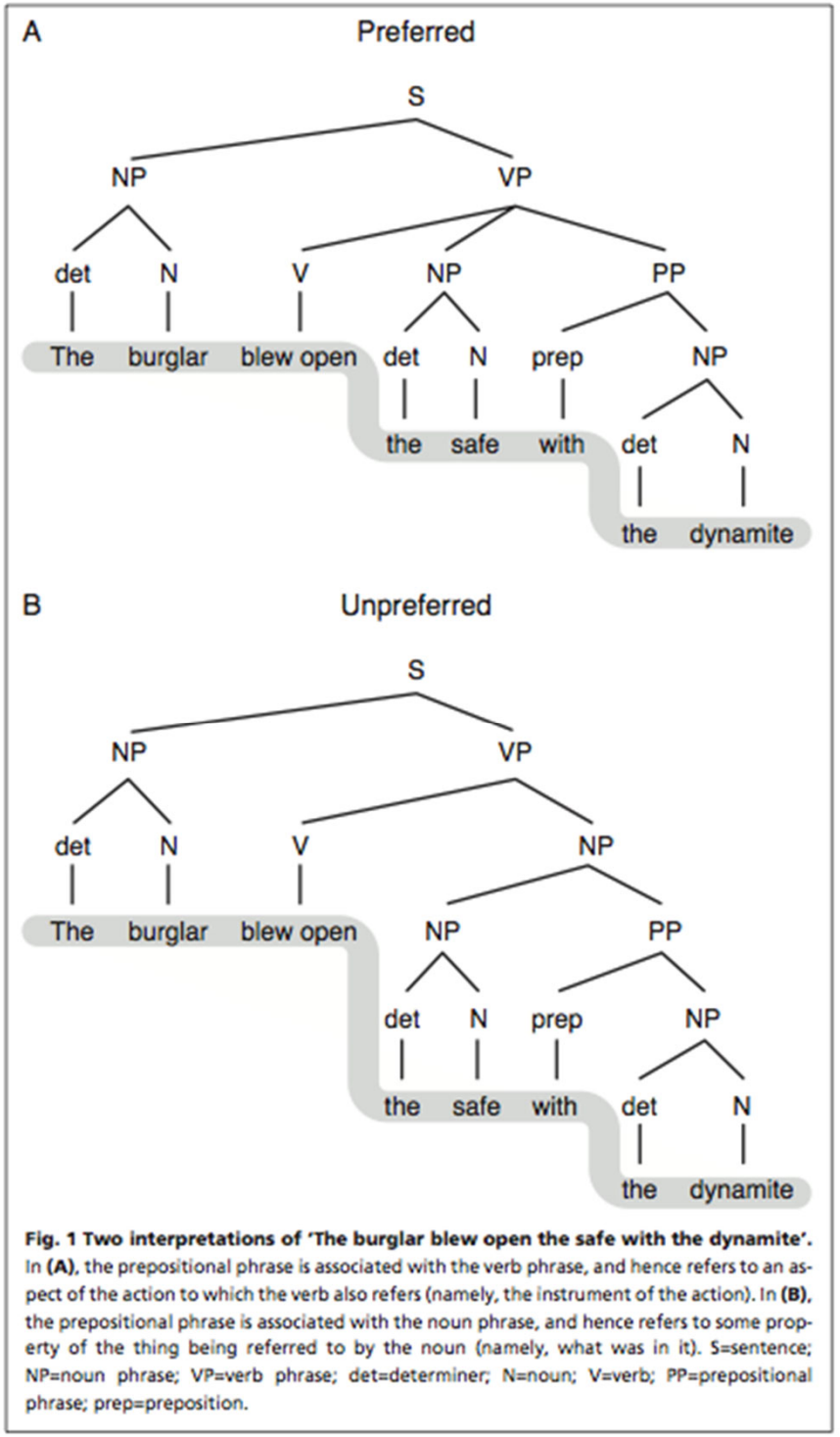


Fig. 1 Two interpretations of ‘The burglar blew open the safe with the dynamite’. In **(A)**, the prepositional phrase is associated with the verb phrase, and hence refers to an aspect of the action to which the verb also refers (namely, the instrument of the action). In **(B)**, the prepositional phrase is associated with the noun phrase, and hence refers to some property of the thing being referred to by the noun (namely, what was in it). S=sentence; NP=noun phrase; VP=verb phrase; det=determiner; N=noun; V=verb; PP=prepositional phrase; prep=preposition.

Frazier, L. (1987) Sentence processing: a tutorial review, in *Attentionand Performance XII: The Psychology of Reading* (Coltheart, M., ed.),pp.

559–586, Erlbaum

Syntactic explanations of the “garden path” effect

We can see late closure at work if we consider a sentence that requires the “unpreferred” parse to be sensible:

“The burglar blew open the safe with the jewels.”

In careful tests of reading time and eye movements, we can see that people are slower to read sentences where the sensible reading requires the unpreferred parse.

That is, you momentarily entertain the idea that the jewels were used to blow open the safe, and then have to retrace your steps.

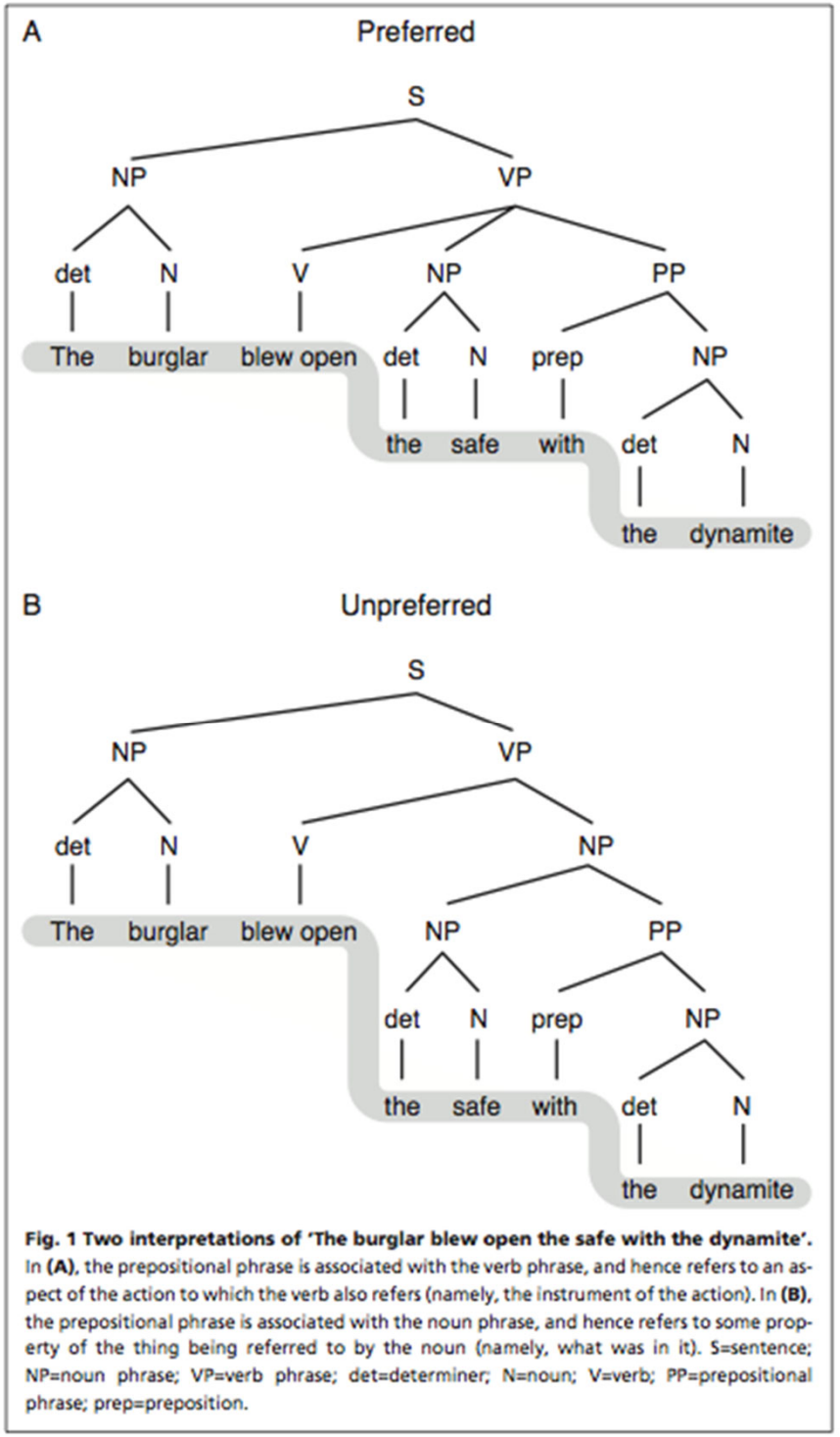


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Syntactic explanations of the “garden path” effect

The explanation rests on the idea of a parser that deals exclusively with the abstract syntactic structure of the sentence (remember the “lumber mill” example from speech perception).

The parser has its own operating characteristics. It wants to make the simplest tree possible (minimal attachment) and it accomplishes this, in part, by leaving clauses open as long as possible (late closure).

This is one way of dealing with ambiguity very *efficiently* – ignore it until it becomes a problem!

Some problems with a purely syntactic explanation

Syntactically "difficult" sentences aren't always difficult to parse.

Table 1
Examples of Sentences With Reduced Relatives

Example	Source
Thus, a noun phrase FOLLOWED BY A MORPHOLOGICALLY AMBIGUOUS VERB (e.g., "The defendant examined") will be temporarily ambiguous.	Trueswell, Tanenhaus, & Garnsey, 1994, p. 287
Referential information PROVIDED BY THE DISCOURSE is of no help.	Britt, Perfetti, Garrod, & Rayner, 1992, p. 305
Recent research REPORTED BY FERREIRA AND CLIFTON (1986) has demonstrated that syntactic processing is quite independent and that the initial syntactic analysis ASSIGNED TO A SENTENCE is little influenced by the semantic information ALREADY ANALYZED.	Frazier & Rayner, 1987, pp. 520–521
The pattern of results OBTAINED WITH GLOBAL READING TIMES is reflected by differences LOCATED ON THE CRITICAL DISAMBIGUATING PHRASE.	Altmann & Steedman, 1988, p. 226
A person PRESENTED WITH *THE EVIDENCE EXAMINED BY THE LAWYER* may interpret *the evidence* as a theme, not an agent, because *examined* requires an animate agent.	Ferreira & Clifton, 1986, p. 350
In all cases, the examples CITED HERE were not the only reduced relatives in these articles.	See table note

Reduced relatives appear in uppercase letters.
Note. In all cases, the examples cited here were not the only reduced relatives in these articles; the rate of reduced relative usage appears to be about one per journal page in this sample of articles.

In all of these sentences, the verbs in ALL CAPS introduce a "reduced relative" clause, as in "the horse raced past the barn fell."

But these are not impossibly difficult sentences.

The alternative argument advanced by MacDonald, Perlmutter & Seidenberg (1994), is that semantic factors and statistical regularities interact with "parsing" from very early on.

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This is a wonderfully passive-aggressive table, BTW:

Many of the examples are drawn from papers that use the difficulty of parsing reduced relatives as evidence for an autonomous parser.

The argument goes that, when we see “Recent research reported by…”

- the fact that “research” is an inanimate object, and unlikely to be the subject of a verb like “report” (semantics)

-The fact that “reported” occurs frequently in reduced relatives (lexical statistics)

- the fact that “reported” occurs very frequently in reduced relatives in scientific papers, especially when following the word “research” (lexical statistics and “world knowledge”)

…and many other factors *interact* with the syntax of the sentence very early on.

Consider:

The burglar arrested in the parking lot was charged today.

The policeman arrested in the parking lot was charged today.

Why?

“World knowledge” can be very local

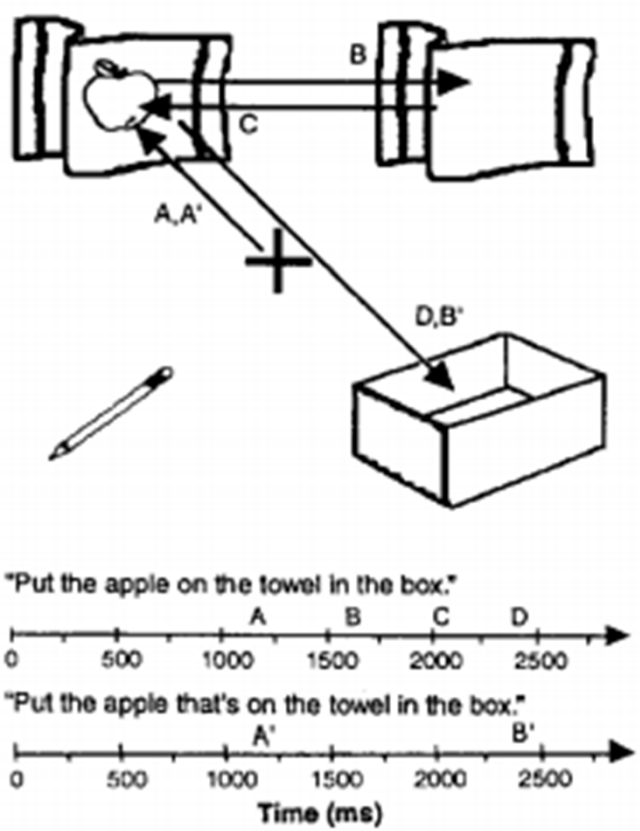


Fig. 1. Typical sequence of eye movements in the one-referent context for the ambiguous and unambiguous instructions. Letters show when in the instruction each eye movement occurred, as determined by the mean latency for that type of eye movement (A′ and B′ correspond to the unambiguous instruction).

Spivey et al. (1995) gave people a visual world with a pencil, a box, a towel and an *apple on a towel.*

People were then instructed to “put the apple on the towel in the box.”

You can imagine that with minimal attachment, “the towel” would temporarily seem like it should be the end of the clause “put the apple on the towel.”

And lo, people do seem to look at the towel while planning their movements, as though they are considering it as a target, before planning a movement to the box.

“World knowledge” can be very local

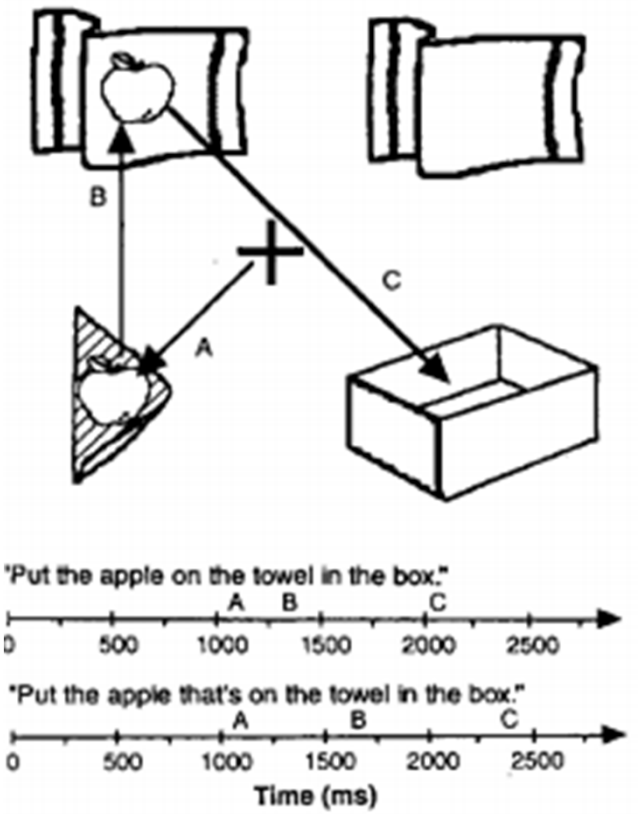


Fig. 2. Typical sequence of eye movements in the two-referent context. Note that the sequence and the timing of eye movements, relative to the nouns in the speech stream, did not differ for the ambiguous and unambiguous instructions.

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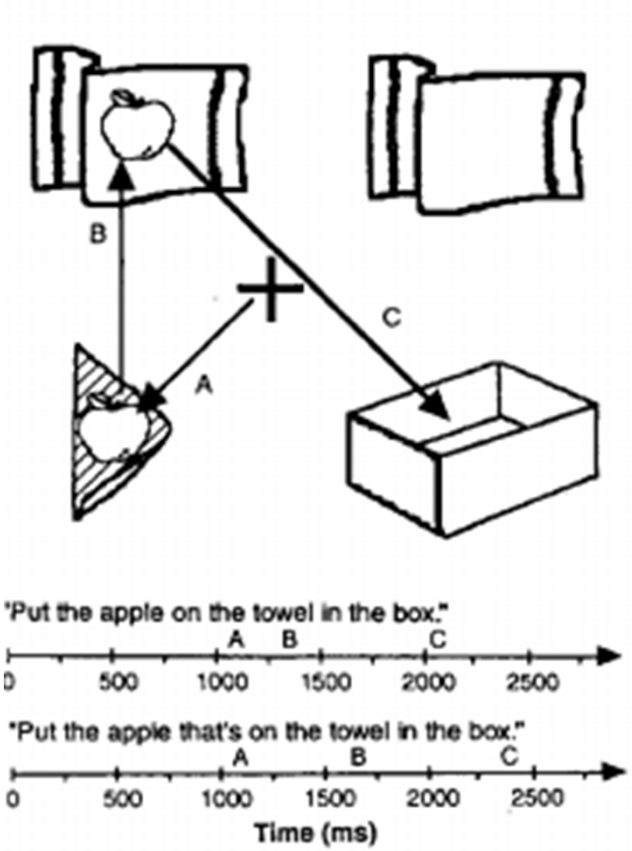


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Now consider what happens when there are *two apples* in the display.

All of a sudden it is *felicitous* to provide more information regarding *which apple you’re talking about.*

So people don’t look at the towel as much.

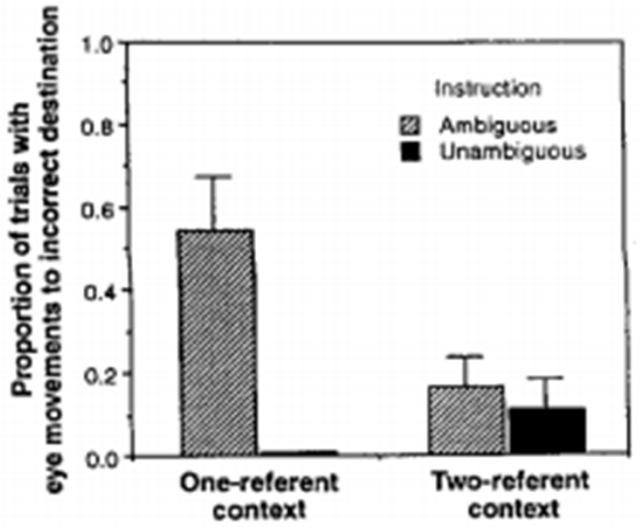


Fig. 3. Proportion of trials in which participants looked at the incorrect destination.

“World knowledge” can be very local

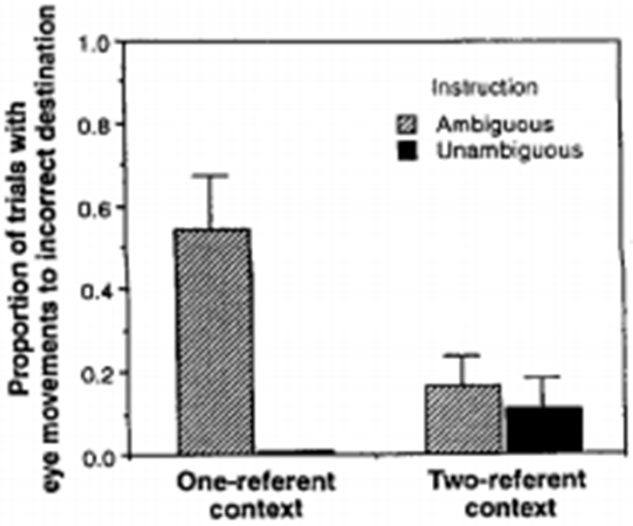
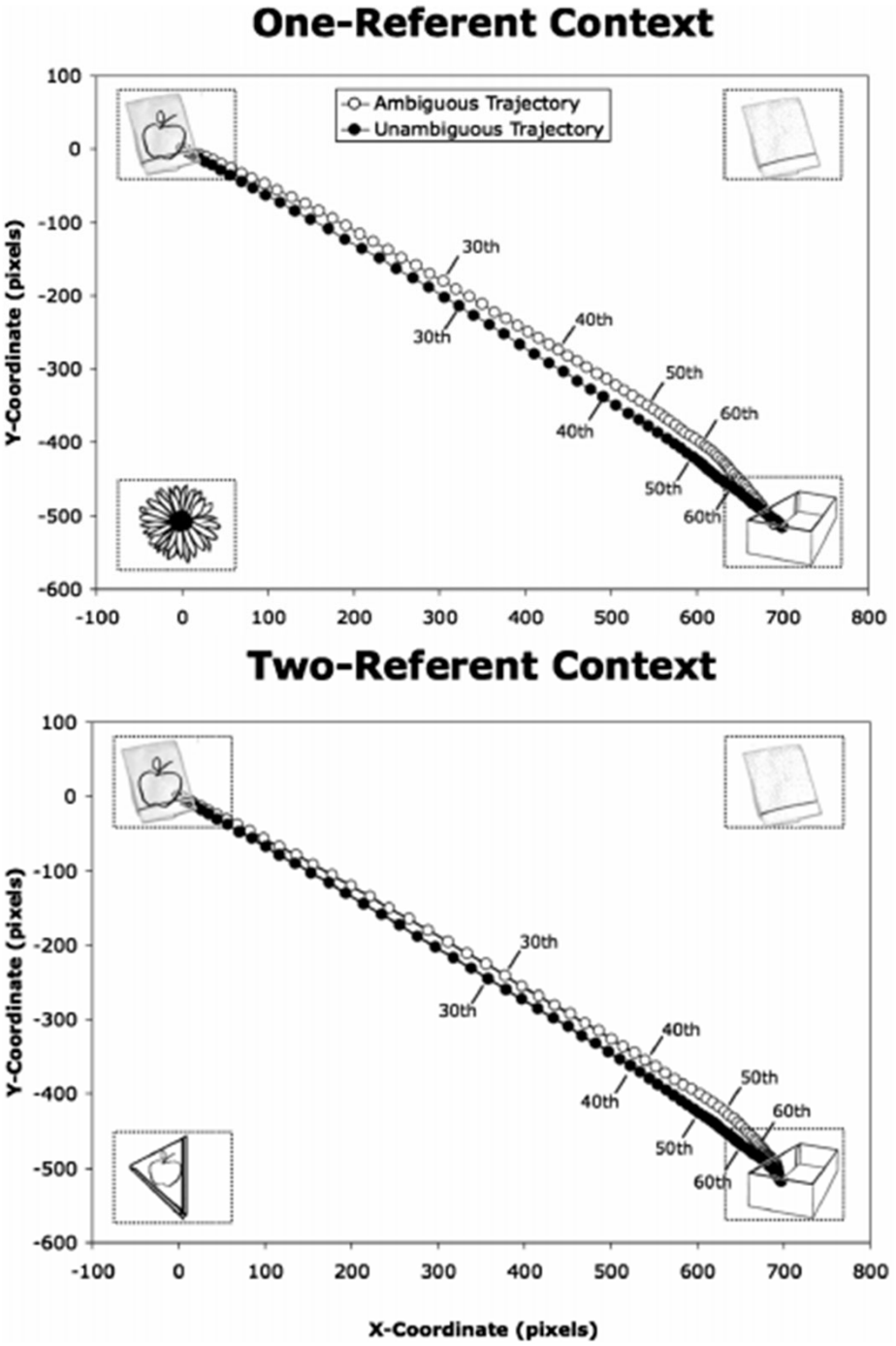


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This is a big problem for purely syntactic accounts.

Different parses of the same sentence seem to be favored under different semantic/pragmatic conditions, and this effect happens very early.



More recently, researchers have begun using data from mouse movements to look at these questions.

It's not just eye-movements, but the actual execution of arm movements that can be influenced by syntactic ambiguity and its interaction with the visually-defined context.

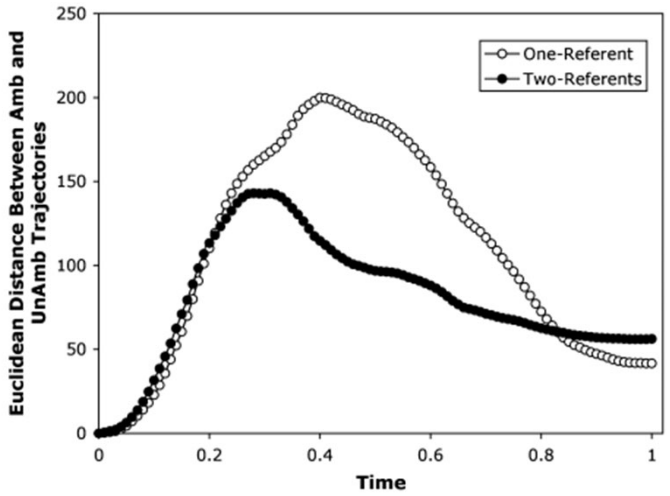


Fig. 2. The Euclidean distance between the ambiguous- and unambiguous-sentence conditions, per context.

Psychophysiological investigations of syntax.

The

pizza

was

in

the

eaten

Psychophysiological investigations of syntax.

The early left anterior negativity (ELAN) is an ERP phenomenon elicited for *word class* violations.

Some classic ELAN stimuli:

The pizza was in the *eaten.*

The scientist criticized Max's *of* proof the theorem.

Features of the ELAN:

- early (100-200ms)
- left/anterior (language-specific?)
- specifically for *word category* violations, i.e., a noun when a verb is expected, or a preposition.

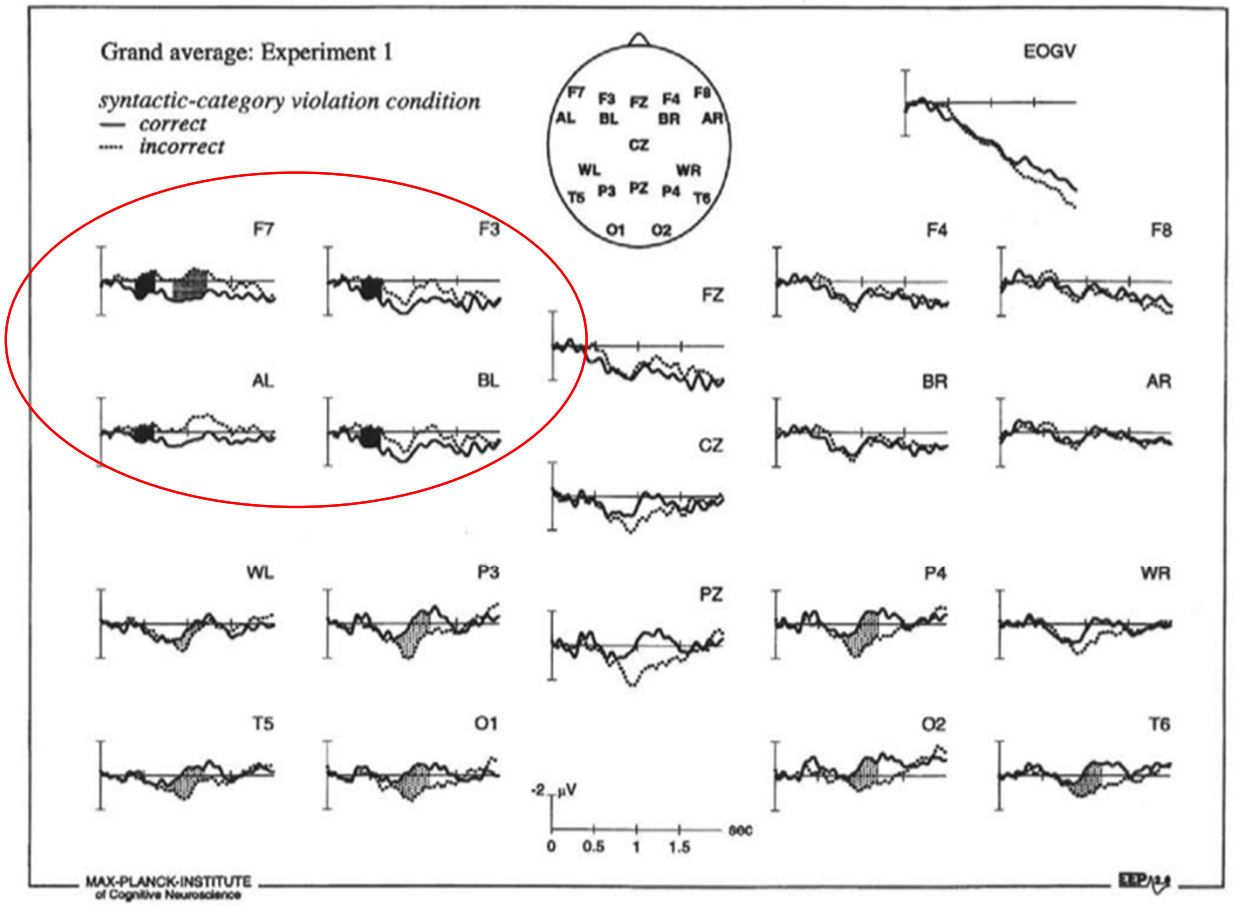


Figure 2. Grand average event-related brain potentials at the electrode sites in the four quadrants and three midline positions elicited by the critical word in the syntactic-category violation condition in Experiment 1. The waveforms are superimposed for correct and incorrect sentences. The ordinate indicates the onset of the critical word. The black shaded area indicates the early left anterior negativity, the gray shaded area a left frontal negativity and the dotted area the posterior positivity. Midline electrodes are excluded from shading as these electrodes were not included in the analyses of variance calculated separately for each quadrant.

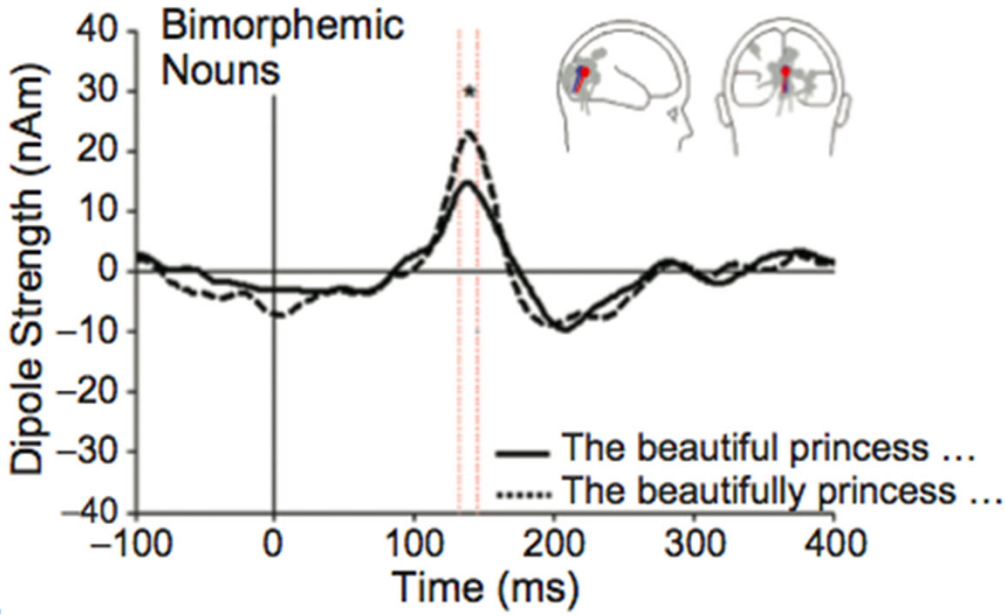
The ELAN*: It came from visual cortex* (with visually presented words).

This is interesting.

It suggests that, when we're expecting a verb (for example), that prediction can feed back down to the earliest parts of the visual system, and predict certain visual patterns that are consistent with the way verbs are written.

The very early response to the error might reflect a mismatch between that predictive process and what finally appears on the screen.

Question: Does this only work for items that are morphologically *marked* as nouns or verbs? Or for open vs. closed class words?



Dikker, S., Rabagliati, H., Farmer, T. A., & Pylkkänen, L. (2010). Early occipital sensitivity to syntactic category is based on form typicality. *Psychological Science 121,* 629-34.

Psychophysiological investigations of syntax.

(3a) Chris and Ben are glad that the bird *perches seem* easy to install.

(3b) Chris and Ben are glad that the bird *perches comfortably* in the cage.

(4a) The teacher told the principal that the student *needs were* not being met.

(4b) The teacher told the principal that the student *needs to be* more focused.

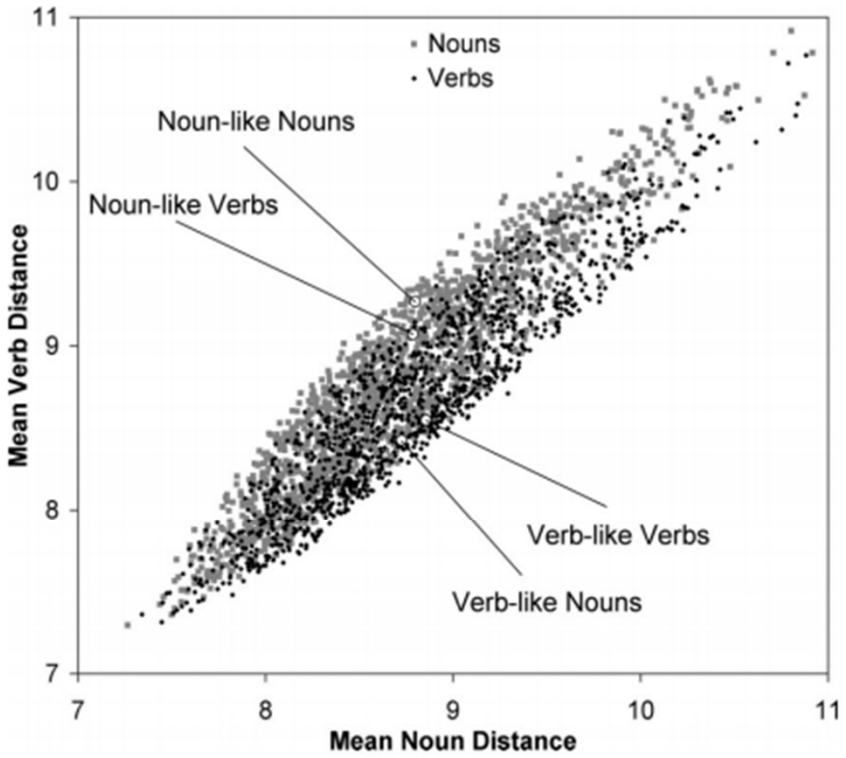


Fig. 1. The 3,158 words from the corpus analyses in experiment 1, plotted as a function of their mean Euclidian distance in phonological feature space to all nouns (*x* axis) and all verbs (*y* axis). Nouns (gray squares) tend to cluster in the upper left and the verbs (black diamonds) in the lower right. The points labeled Noun-like Nouns and Verb-like Nouns indicate the center of the phonologically typical and atypical nouns, respectively, used in experiment 2. Similarly, the points Verb-like Verbs and Noun-like Verbs denote the center of the typical and atypical verbs used in experiment 3.

Psychophysiological investigations of syntax.

(3a) Chris and Ben are glad that the bird *perches seem* easy to install.

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(4a) The teacher told the principal that the student *needs were* not being met.

(4b) The teacher told the principal that the student *needs to be* more focused.

[BTW “perches” is noun-like, and “needs” is verb-like]

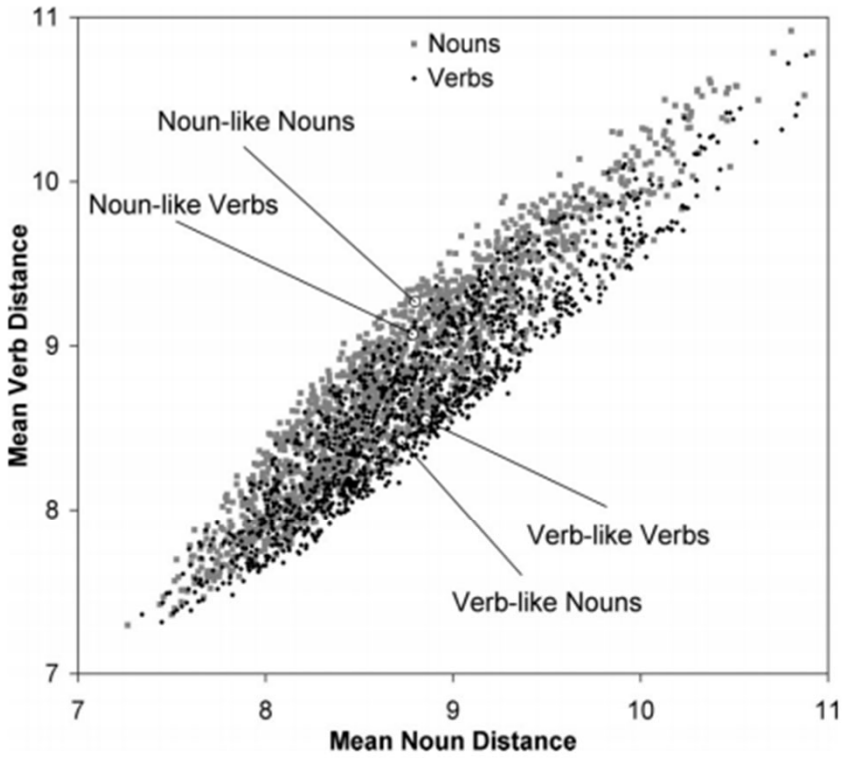


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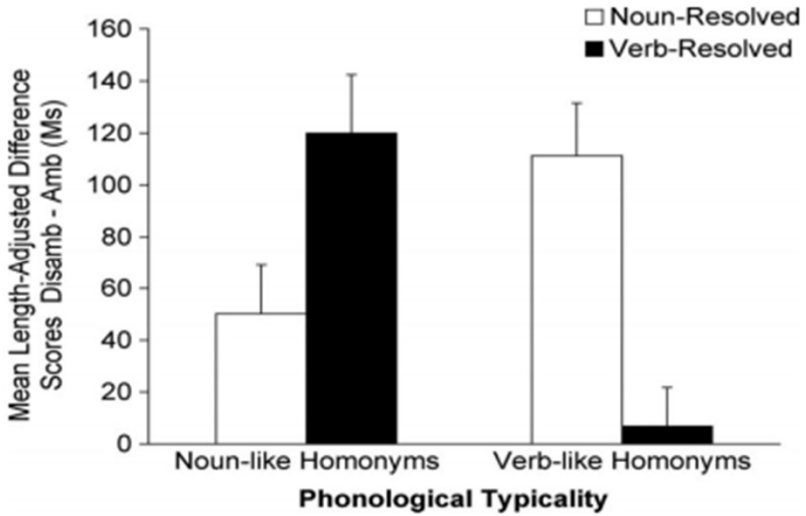


Fig. 3. Mean difference (disambiguation minus point of ambiguity) scores (and standard errors) for each of the four possible conditions in experiment 4. Rising bars indicate that RTs increased from the point of ambiguity to disambiguation.

Psychophysiological investigations of syntax.

The ELAN effect for nouns where a verb is expected is limited to those cases where there is either syntactic (i.e., morphemes indicating noun-ness) or statistical form-based information (i.e., “nouniness”) *physically available*.

No effect was observed for neutral nouns like “infant” that lack strong cues.

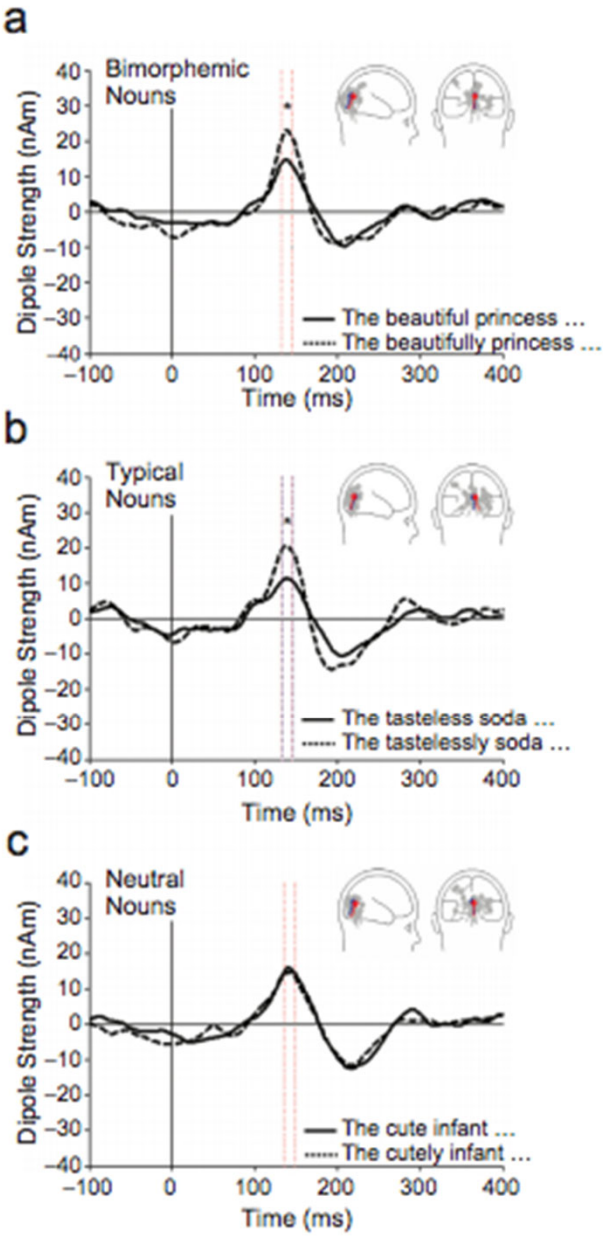


Fig. 1. Grand-averaged waveforms for the M100 dipole sources for expected and unexpected (a) bimorphemic, (b) typical, and (c) neutral nouns ($n = 15$). The dotted vertical lines indicate 15-ms intervals centered around the average M100 peak. Mean dipole locations and orientations (blue for expected nouns, red for unexpected nouns), as well as the dipoles from the individual participants (gray), are plotted for each noun type. Asterisks indicate a significant difference in M100 amplitude between unexpected and expected nouns (*$p < .05$).

Sentence processing in four sentences.

Colorless green ideas sleep furiously.

Syntax is independent of semantics to some extent. In the limit you can have sentences that are grammatically correct but meaningless.

The horse raced past the barn fell.

The syntactic structure of a sentence is often highly ambiguous (actually *vague*), but we don't notice except where things fail. Reading times for "preferred" vs. "unpreferred" structures suggest that syntactic processes take shortcuts to save memory, rather than activating every possible parse of a sentence.

Put the apple on the towel in the box.

Context, pragmatics, and other factors seem to interact with syntactic processing from very early on.

The bird perches were comfortable.

The very earliest syntactic effects observable in the brain are driven in part by form typicality, not necessarily by morphosyntactic structures, and seem to be related to predictions about the *physical forms of words.*

For next class

Read Menn, Chapter 6 (on aphasias)